

Easy way to remember
EIGRP = "Router learns the best route automatically."

Search Labs

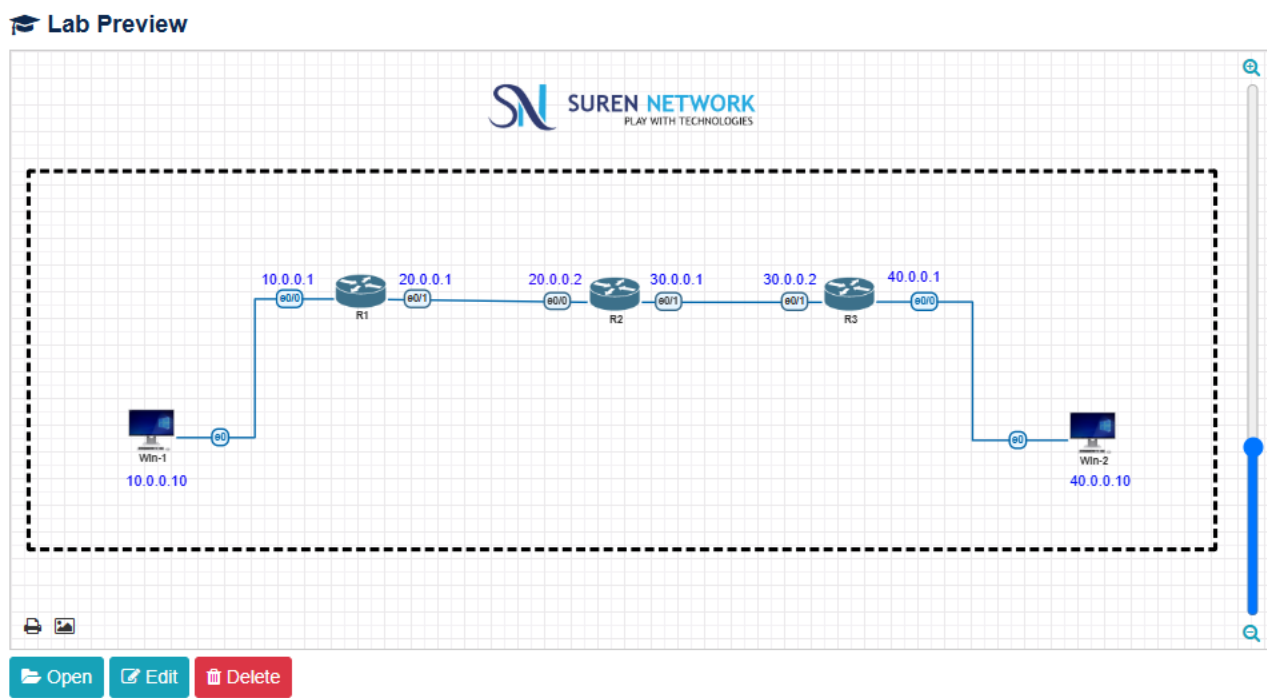
OSPF Area.unl 01 Sep 2026 08:15

4-LAB OSPF PART-1.unl 04 Sep 2025 09:38

3-LAB OSPF DR BDR DROTHERS.unl 04 Sep 2025 09:38

☒ 2-LAB EIGRP.unl Move to Rename Clone

1-LAB RIP.unl 04 Sep 2025 09:38



EIGRP — Enhanced Interior Gateway Routing Protocol

EIGRP is a Cisco-developed dynamic routing protocol used by routers to automatically learn and select the best paths to destination networks.

Important EIGRP terms

1. **Neighbor** → Router directly connected and exchanging EIGRP information.
2. **Successor** → The **best route** to a destination.
3. **Feasible Successor** → A **backup route** that satisfies EIGRP's feasibility condition.
4. **DUAL** → Calculates loop-free paths and helps EIGRP converge quickly when a route fails.

Lab Name : 2-LAB EIGRP

Author : 1

Version : 1

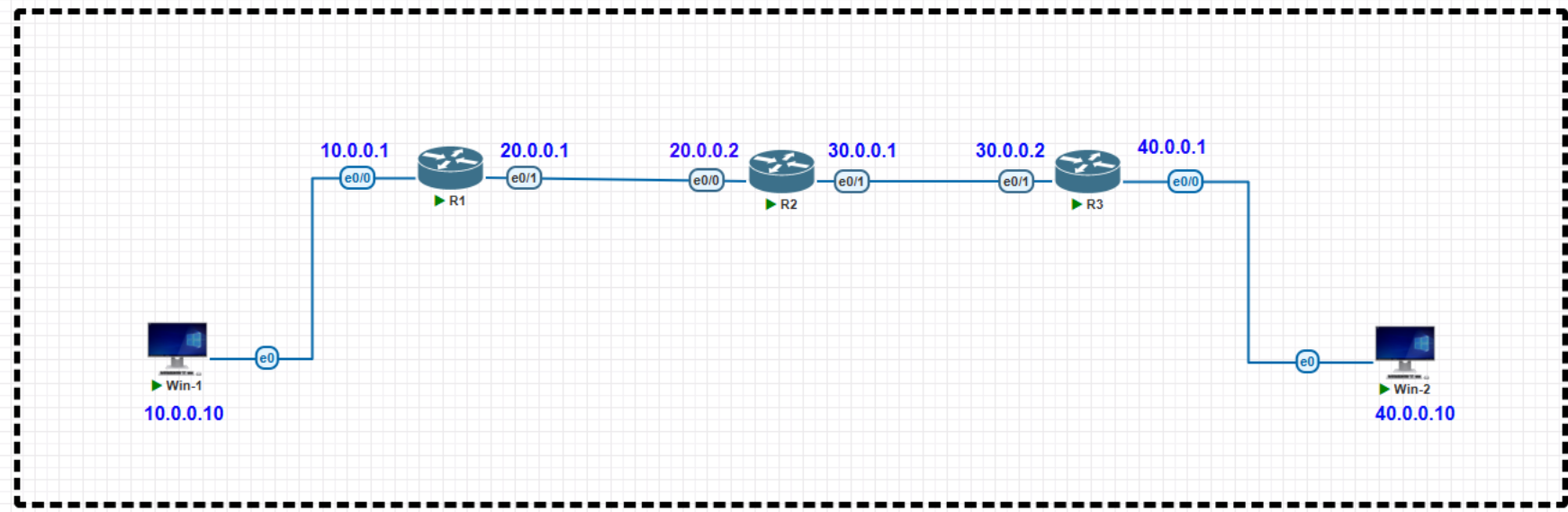
Lab ID : 6ac83e3e-4a2a-4b1e-8c67-7d3712068df3

Description :

Title: EIGRP

LAB:- II

By Meganadhan M A



PNETLab | Topology x Terminal x lab eigrp x acd x GRE.cdr x nat x

labs.surennetwork.com/store/public/admin/labs/terminal?terminals=eyJ0YWJzljpbeyJuYW1lIjoV2luLTEiLCJ1cmwiOiIvaHRtbDUvly9jbGllbnQvTXpBNE5qUXINREk0QUd

PC1 IP assigned manually

Win-1 x R1 x R2 x R3 x Win-2 x

Network Connections

File Edit View Favorites Tools Advanced Help

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Address Network Connections

Network Tasks

- Create a new connection
- Set up a home or small office network
- Change Windows Firewall settings
- Disable this network device
- Repair this connection
- Rename this connection
- View status of this connection
- Change settings of this connection

Other Places

- Control Panel
- My Network Places
- My Documents
- My Computer

Details

Local Area Connection
LAN or High-Speed Internet
Connected
Intel(R) PRO/1000 MT Network Connection
IP Address: 10.0.0.10
Subnet Mask: 255.0.0.0
Manually Configured

LAN or High-Speed Internet

Local Area Connection
Connected
Intel(R) PRO/1000 MT Network Connection

Local Area Connection Status

General Support

Connection status

Address Type:	Manually Configured
IP Address:	10.0.0.10
Subnet Mask:	255.0.0.0
Default Gateway:	10.0.0.1

Details...

Windows did not detect problems with this connection. If you cannot connect, click Repair.

Repair

Close

Start

33°C Mostly cloudy

Search

C:\WINDOWS\system32... Network Connections Local Area

Terminal x lab eigrp x acd x GRE.cdr x nat x

network.com/store/public/admin/labs/terminal?terminals=eyJ0YWJzljpbeyJuYW1lIjoV2luLTEiLCJ1cmwiOiIvaHRtbDUvly9jbGllbnQvTXpBNE5qUXINREk0QUd

PC2 IP assigned manually

Win-2 x

Network Connections

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Address Network Connections

Network Tasks

- Create a new connection
- Set up a home or small office network
- Change Windows Firewall settings
- Disable this network device
- Repair this connection
- Rename this connection
- View status of this connection
- Change settings of this connection

Other Places

- Control Panel
- My Network Places
- My Documents
- My Computer

Details

Local Area Connection
LAN or High-Speed Internet

LAN or High-Speed Internet

Local Area Connection
Connected
Intel(R) PRO/1000 MT Network Connection

Local Area Connection Status

General Support

Connection status

Address Type:	Manually Configured
IP Address:	40.0.0.10
Subnet Mask:	255.0.0.0
Default Gateway:	40.0.0.1

Details...

Windows did not detect problems with this connection. If you cannot connect, click Repair.

Repair

Close

Start

33°C Mostly cloudy

Search

C:\WINDOWS\system32... Network Connections Local Area

```
Win-1 x R1 x R2 x R3 x Win-2 x

R1#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              10.0.0.1        YES TFTP    up          up
Ethernet0/1              20.0.0.1        YES TFTP    up          up
Ethernet0/2              unassigned      YES TFTP    down        down
Ethernet0/3              unassigned      YES TFTP    down        down
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#router eigrp 1
R1(config-router)#network 10.0.0.0
R1(config-router)#network 20.0.0.0
R1(config-router)#e
% Ambiguous command: "e"
R1(config-router)#ex
R1(config)#
*Sep 22 07:39:28.079: %DUAL-5-NBRCHANGE: EIGRP-IPv4 1: Neighbor 20.0.0.2 (Ethernet0/1) is up: new adjacency
R1(config)#
*Sep 22 07:40:24.140: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1(config)#
R1(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/8 is directly connected, Ethernet0/0
L    10.0.0.1/32 is directly connected, Ethernet0/0
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/1
L    20.0.0.1/32 is directly connected, Ethernet0/1
D    30.0.0.0/8 [90/307200] via 20.0.0.2, 01:17:52, Ethernet0/1
D    40.0.0.0/8 [90/332800] via 20.0.0.2, 01:16:44, Ethernet0/1
R1(config)#
```

→ To check the interface Assigned network

→ To enable the EIGRP Protocol cmd, 1 is the EIGRP group

→ Here, D means EIGRP Protocol

→ The routes are automatically assigned by the EIGRP tool. These all Will run in the background only.

Router (R2) Configuration

Win-1 x R1 x R2 x R3 x Win-2 x

R2#sh ip int br

Interface

IP-Address

OK? Method Status

Protocol

Ethernet0/0

20.0.0.2

YES TFTP up

up

Ethernet0/1

30.0.0.1

YES TFTP up

up

Ethernet0/2

unassigned

YES TFTP down

down

Ethernet0/3

unassigned

YES TFTP down

down

R2#

R2#config t

Enter configuration commands, one per line. End with CNTL/Z.

R2(config)#router eigrp 1

R2(config-router)#network 20.0.0.0

R2(config-router)#network 20.0.0.0

*Sep 22 07:39:28.084: %DUAL-5-NBRCHANGE: EIGRP-IPv4 1: Neighbor 20.0.0.1 (Ethernet0/0) is up: new adjacency

R2(config-router)#network 30.0.0.0

R2(config-router)#ex

R2(config)#

*Sep 22 07:40:06.264: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)

R2(config)#

*Sep 22 07:40:34.056: %DUAL-5-NBRCHANGE: EIGRP-IPv4 1: Neighbor 30.0.0.2 (Ethernet0/1) is up: new adjacency

R2(config)#

R2(config)#do sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

D 10.0.0.0/8 [90/307200] via 20.0.0.1, 01:18:15, Ethernet0/0

20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 20.0.0.0/8 is directly connected, Ethernet0/0

L 20.0.0.2/32 is directly connected, Ethernet0/0

30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 30.0.0.0/8 is directly connected, Ethernet0/1

L 30.0.0.1/32 is directly connected, Ethernet0/1

D 40.0.0.0/8 [90/307200] via 30.0.0.2, 01:17:04, Ethernet0/1

R2(config)#

Router (R3) Configuration

Win-1 x R1 x R2 x R3 x Win-2 x

Interface

IP-Address

OK? Method Status

Protocol

Ethernet0/0

40.0.0.1

YES TFTP up

up

Ethernet0/1

30.0.0.2

YES TFTP up

up

Ethernet0/2

unassigned

YES TFTP down

down

Ethernet0/3

unassigned

YES TFTP down

down

R3#

R3#config t

Enter configuration commands, one per line. End with CNTL/Z.

R3(config)#

R3(config)#router e

*Sep 22 07:40:16.768: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)

R3(config)#router eigrp 1

R3(config-router)#network 30.0.0.0

R3(config-router)#network 30.0.0.0

*Sep 22 07:40:34.061: %DUAL-5-NBRCHANGE: EIGRP-IPv4 1: Neighbor 30.0.0.1 (Ethernet0/1) is up: new adjacency

R3(config-router)#network 40.0.0.0

R3(config-router)#ex

R3(config)#

R3(config)#do sh ip int br

Interface

IP-Address

OK? Method Status

Protocol

Ethernet0/0

40.0.0.1

YES TFTP up

up

Ethernet0/1

30.0.0.2

YES TFTP up

up

Ethernet0/2

unassigned

YES TFTP down

down

Ethernet0/3

unassigned

YES TFTP down

down

R3(config)#do sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

D 10.0.0.0/8 [90/332800] via 30.0.0.1, 01:17:26, Ethernet0/1

D 20.0.0.0/8 [90/307200] via 30.0.0.1, 01:17:26, Ethernet0/1

30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 30.0.0.0/8 is directly connected, Ethernet0/1

L 30.0.0.2/32 is directly connected, Ethernet0/1

40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 40.0.0.0/8 is directly connected, Ethernet0/0

L 40.0.0.1/32 is directly connected, Ethernet0/0

R3(config)#

Win-1 x R1 x R2 x R3 x Win-2 x

```
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/1
L    20.0.0.1/32 is directly connected, Ethernet0/1
D    30.0.0.0/8 [90/307200] via 20.0.0.2, 01:17:52, Ethernet0/1
D    40.0.0.0/8 [90/332800] via 20.0.0.2, 01:16:44, Ethernet0/1
R1(config)#
R1(config)#router eigrp ?
<1-65535> Autonomous System
WORD      EIGRP Virtual-Instance Name

R1(config)#do sh ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(1)
H   Address           Interface           Hold Uptime    SRTT   RT0  Q  Seq
                               (sec)          (ms)          Cnt  Num
0   20.0.0.2           Et0/1              12 01:27:07     1    100 0   7

R1(config)#
R1(config)#do sh int e0/0
Ethernet0/0 is up, line protocol is up
Hardware is AmdP2, address is aabb.cc00.0300 (bia aabb.cc00.0300)
Internet address is 10.0.0.1/8
MTU 1500 bytes, BW 10000 Kbit/sec, DLY 1000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Auto-duplex, Auto Speed, media type is unknown media type
input flow-control is off, output flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:05:20, output 00:00:00, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
  135 packets input, 21497 bytes, 0 no buffer
    Received 84 broadcasts (0 IP multicasts)
      0 runs, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
    0 input packets with dribble condition detected
 1926 packets output, 169150 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 unknown protocol drops
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out
R1(config)#
```

To checking neighbour route or gateway

Interface connection

News for you
IIT Bombay stud...

Search

02:39 PM
22-09-2026

Same as the previous slide

```
Win-1 x R1 x R2 x R3 x Win-2 x
D 10.0.0.0/8 [90/307200] via 20.0.0.1, 01:18:15, Ethernet0/0
D 20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 20.0.0.0/8 is directly connected, Ethernet0/0
L 20.0.0.2/32 is directly connected, Ethernet0/0
C 30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 30.0.0.0/8 is directly connected, Ethernet0/1
L 30.0.0.1/32 is directly connected, Ethernet0/1
D 40.0.0.0/8 [90/307200] via 30.0.0.2, 01:17:04, Ethernet0/1
R2(config)#
R2(config)#do sh ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(1)
H Address Interface Hold Uptime SRTT RT0 Q Seq
(sec) (ms) Cnt Num
1 30.0.0.2 Et0/1 13 01:26:32 4 100 0 3
0 20.0.0.1 Et0/0 12 01:27:38 4 100 0 4
R2(config)#
R2(config)#do sh int e0/0
Ethernet0/0 is up, line protocol is up
Hardware is AmdP2, address is aabb.cc00.0100 (bia aabb.cc00.0100)
Internet address is 20.0.0.2/8
MTU 1500 bytes, BW 10000 Kbit/sec, DLY 1000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Auto-duplex, Auto Speed, media type is unknown media type
input flow-control is off, output flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:03, output 00:00:00, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
1335 packets input, 133928 bytes, 0 no buffer
Received 1323 broadcasts (0 IP multicasts)
0 runs, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
0 input packets with dribble condition detected
1936 packets output, 170167 bytes, 0 underruns
0 output errors, 0 collisions, 1 interface resets
0 unknown protocol drops
0 babbles, 0 late collision, 0 deferred
0 lost carrier, 0 no carrier
0 output buffer failures, 0 output buffers swapped out
R2(config)#
```

Same as the previous slide

```
Win-1 x R1 x R2 x R3 x Win-2 x
D 10.0.0.0/8 [90/332800] via 30.0.0.1, 01:17:26, Ethernet0/1
D 20.0.0.0/8 [90/307200] via 30.0.0.1, 01:17:26, Ethernet0/1
C 30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 30.0.0.0/8 is directly connected, Ethernet0/1
L 30.0.0.2/32 is directly connected, Ethernet0/1
C 40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 40.0.0.0/8 is directly connected, Ethernet0/0
L 40.0.0.1/32 is directly connected, Ethernet0/0
R3(config)#
R3(config)#do sh ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(1)
H Address Interface Hold Uptime SRTT RT0 Q Seq
(sec) (ms) Cnt Num
0 30.0.0.1 Et0/1 14 01:26:50 1600 5000 0 6
R3(config)#
R3(config)#do sh int e0/0
Ethernet0/0 is up, line protocol is up
Hardware is AmdP2, address is aabb.cc00.0200 (bia aabb.cc00.0200)
Internet address is 40.0.0.1/8
MTU 1500 bytes, BW 10000 Kbit/sec, DLY 1000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Auto-duplex, Auto Speed, media type is unknown media type
input flow-control is off, output flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:04:24, output 00:00:03, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
139 packets input, 22773 bytes, 0 no buffer
Received 85 broadcasts (0 IP multicasts)
0 runs, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
0 input packets with dribble condition detected
1919 packets output, 168884 bytes, 0 underruns
0 output errors, 0 collisions, 1 interface resets
0 unknown protocol drops
0 babbles, 0 late collision, 0 deferred
0 lost carrier, 0 no carrier
0 output buffer failures, 0 output buffers swapped out
R3(config)#
```

```
R1(config)#do sh ip eigrp topology
EIGRP-IPv4 Topology Table for AS(1)/ID(20.0.0.1)
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - reply Status, s - sia Status
```

```
P 10.0.0.0/8, 1 successors, FD is 281600
  via Connected, Ethernet0/0
P 30.0.0.0/8, 1 successors, FD is 307200
  via 20.0.0.2 (307200/281600), Ethernet0/1
P 40.0.0.0/8, 1 successors, FD is 332800
  via 20.0.0.2 (332800/307200), Ethernet0/1
P 20.0.0.0/8, 1 successors, FD is 281600
  via Connected, Ethernet0/1
```

Router 1

Router 2

```
R2(config)#do sh ip eigrp topology
EIGRP-IPv4 Topology Table for AS(1)/ID(30.0.0.1)
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - reply Status, s - sia Status
```

```
P 10.0.0.0/8, 1 successors, FD is 307200
  via 20.0.0.1 (307200/281600), Ethernet0/0
P 30.0.0.0/8, 1 successors, FD is 281600
  via Connected, Ethernet0/1
P 40.0.0.0/8, 1 successors, FD is 307200
  via 30.0.0.2 (307200/281600), Ethernet0/1
P 20.0.0.0/8, 1 successors, FD is 281600
  via Connected, Ethernet0/0
```

```
R2(config)#
```

```
R3(config)#do sh ip eigrp topology
EIGRP-IPv4 Topology Table for AS(1)/ID(40.0.0.1)
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - reply Status, s - sia Status
```

```
P 10.0.0.0/8, 1 successors, FD is 332800
  via 30.0.0.1 (332800/307200), Ethernet0/1
P 30.0.0.0/8, 1 successors, FD is 281600
  via Connected, Ethernet0/1
P 40.0.0.0/8, 1 successors, FD is 281600
  via Connected, Ethernet0/0
P 20.0.0.0/8, 1 successors, FD is 307200
  via 30.0.0.1 (307200/281600), Ethernet0/1
```

```
R3(config)#
```

Router 3

PC1 to PC2 reachability checking

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>cd\
C:\>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    Autoconfiguration IP Address. . . : 169.254.212.214
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 

C:\>ncpa.cpl
C:\>ping 40.0.0.10

Pinging 40.0.0.10 with 32 bytes of data:

Reply from 40.0.0.10: bytes=32 time=3ms TTL=125
Reply from 40.0.0.10: bytes=32 time=1ms TTL=125
Reply from 40.0.0.10: bytes=32 time=1ms TTL=125
Reply from 40.0.0.10: bytes=32 time=1ms TTL=125

Ping statistics for 40.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\>_
```

PC2 to PC1 reachability checking

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>cd\
C:\>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    Autoconfiguration IP Address. . . : 169.254.211.141
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 

C:\>ncpa.cpl
C:\>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 40.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 40.0.0.1

C:\>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=1ms TTL=125
Reply from 10.0.0.10: bytes=32 time=1ms TTL=125
Reply from 10.0.0.10: bytes=32 time=2ms TTL=125
Reply from 10.0.0.10: bytes=32 time=1ms TTL=125

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>_
```

EIGRP CONFIGURATION

Feature	EIGRP
Type	Dynamic, advanced distance-vector
Algorithm	DUAL (Diffusing Update Algorithm)
Metric	Bandwidth + Delay by default
Administrative Distance	90 (internal), 170 (external)
Protocol number	88
Updates	Partial & triggered
Convergence	Fast
Supports VLSM/CIDR	✓ Yes
IPv4 / IPv6	✓ Both

Step 1 : Assign ip address in all cisco routers

Step 2 : Check interface parameter

```
R1# show interface ethernet 0/0
```

Step 3 : Enable eigrp protocol in all cisco router

```
R1(config)# router eigrp 10           ( To enable eigrp )
R1(config-router)# network 10.0.0.0   ( To enable eigrp in 10 network interface )
R1(config-router)# network 20.0.0.0   ( To enable eigrp in 20 network interface )
R1(config-router)# exit
R1(config)#
```

```
R2(config)# router eigrp 10           ( To enable eigrp )
R2(config-router)# network 20.0.0.0   ( To enable eigrp in 20 network interface )
R2(config-router)# network 30.0.0.0   ( To enable eigrp in 30 network interface )
R2(config-router)# exit
R2(config)#
```

```
R3(config)# router eigrp 10           ( To enable eigrp )
R3(config-router)# network 30.0.0.0   ( To enable eigrp in 30 network interface )
R3(config-router)# network 40.0.0.0   ( To enable eigrp in 40 network interface )
R3(config-router)# exit
R3(config)#
```

Step 4 : Check eigrp neighbor table

```
R1# show ip eigrp neighbor
```

Step 5 : Check eigrp topology table

```
R1# show ip eigrp topology
```

Step 6 : Check routing table

```
R1# show ip route
```